UNIVERSITY OF ZAMBIA

MSC ARTIFICIAL INTELLIGENCE

CSC6302 - COMPUTER VISION

PRACTICAL 7: DEPLOYED COMPUTER VISION FOR THE REAL WORLD - BUILD, TEST, SHIP

PROJECT TITLE: LANDSCAPE IMAGE CLASSIFICATION USING MOBILENETV3 AND STREAMLIT

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1. Introduction

This capstone project presents a web-based computer vision application that classifies photographs into three categories: buildings, forest and sea. The application uses a fine-tuned MobileNetV3-Small model developed during Practical 4.

The system integrates image preprocessing, classification, model evaluation, automated testing and cloud deployment. Python and PyTorch were used for model inference, Streamlit for the web interface, and GitHub Actions for continuous integration.

2. System Design and Implementation

The application uses a fine-tuned MobileNetV3-Small convolutional neural network developed during Practical 4. Transfer learning was selected because it reduces training requirements by reusing features learned from a larger image dataset.

The system follows four stages:

Image upload: Users submit JPG or PNG photographs through the Streamlit interface.

Preprocessing: Images are converted to RGB, resized to 224 × 224 pixels and normalised using ImageNet statistics.

Classification: MobileNetV3 predicts probabilities for buildings, forest and sea.

Result presentation: The application displays the predicted category, confidence score and a plain-language explanation.

A confidence threshold of 60% identifies uncertain predictions. Basic validation handles missing, invalid or excessively small images.

The application was implemented using Python, PyTorch and Streamlit. The trained model was reused without additional training.

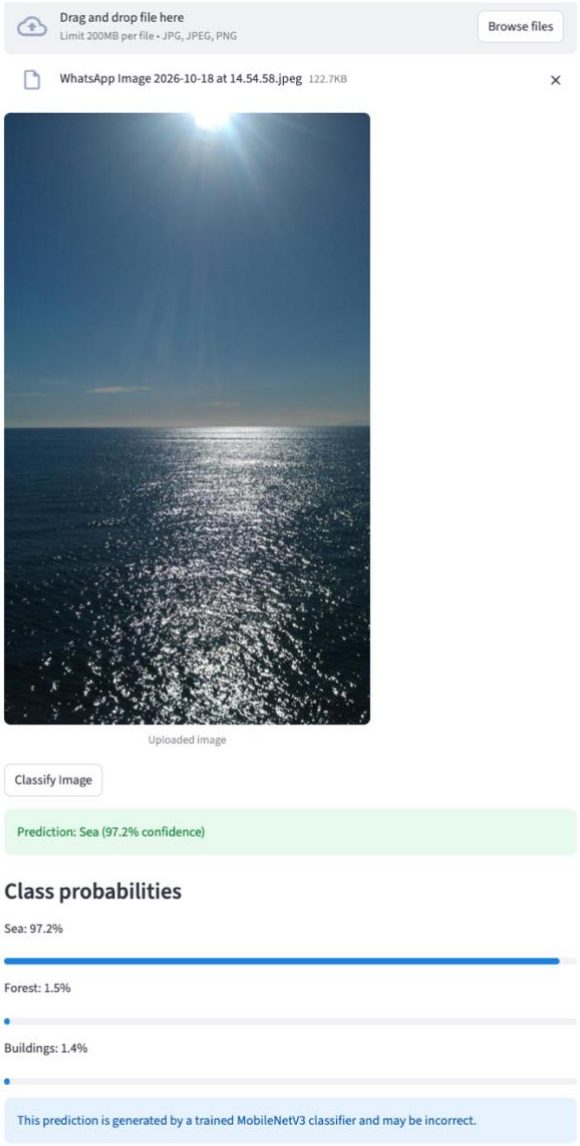


Figure 1: Streamlit image classification interface showing a sea prediction.

3. Model Evaluation and Results

The fine-tuned MobileNetV3-Small model was evaluated using 135 images from the Intel Image Classification test dataset, comprising 45 images each of buildings, forest and sea.

Accuracy, precision, recall and F1-score were calculated using scikit-learn. The model correctly classified all 135 images, achieving 100% accuracy and a macro F1-score of 1.0000.

Table 1: Classification Performance on the Test Dataset

Class	Images	Precision	Recall	F1-score
Buildings	45	1	1	1
Forest	45	1	1	1
Sea	45	1	1	1
Overall	135	1	1	1

The confusion matrix showed no misclassifications within the evaluated dataset. However, this perfect result does not guarantee equivalent performance on unfamiliar photographs. Further independent testing is required to confirm generalisation and rule out potential data leakage.

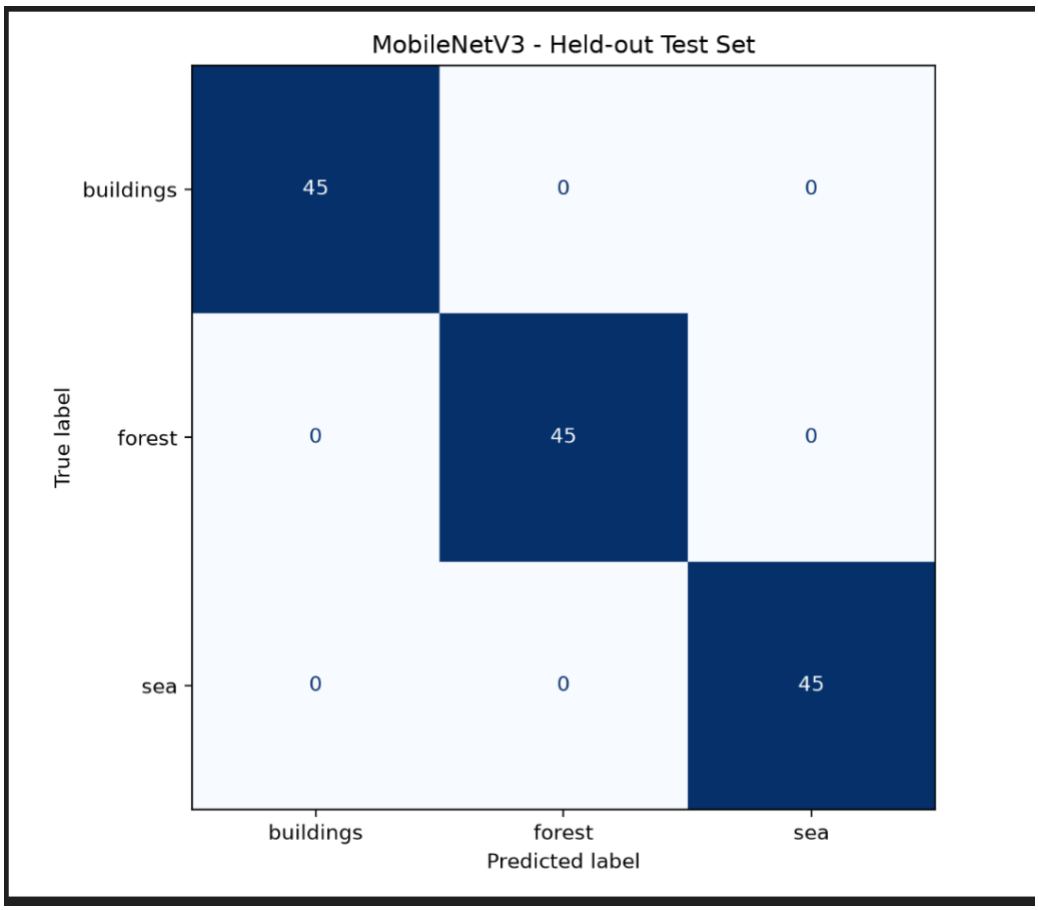


Figure 2: Confusion matrix showing MobileNetV3 classification results for 135 test images.

4. Automated Testing, CI/CD and Deployment

GitHub was used for version control, while GitHub Actions provided continuous integration (CI). Seven automated tests were implemented using pytest to check model loading, preprocessing dimensions, prediction outputs, input validation and probability calculations.

All seven tests passed. Ruff was used for static code analysis, and the GitHub Actions workflow completed successfully.

The application was publicly deployed using Streamlit Community Cloud, connected to the GitHub repository. Updates pushed to the main branch trigger automatic redeployment.

4.1 Deployment Challenges

The original deployment plan used Hugging Face Spaces. However, the platform requested a paid subscription to create a Gradio Space. Streamlit Community Cloud was therefore selected as a free alternative.

The initial Streamlit deployment also failed because the specified PyTorch version was unavailable in the hosting environment. Dependency versions were updated to resolve the installation problem.

These challenges highlighted the importance of dependency compatibility, accessible hosting and alternative deployment strategies.

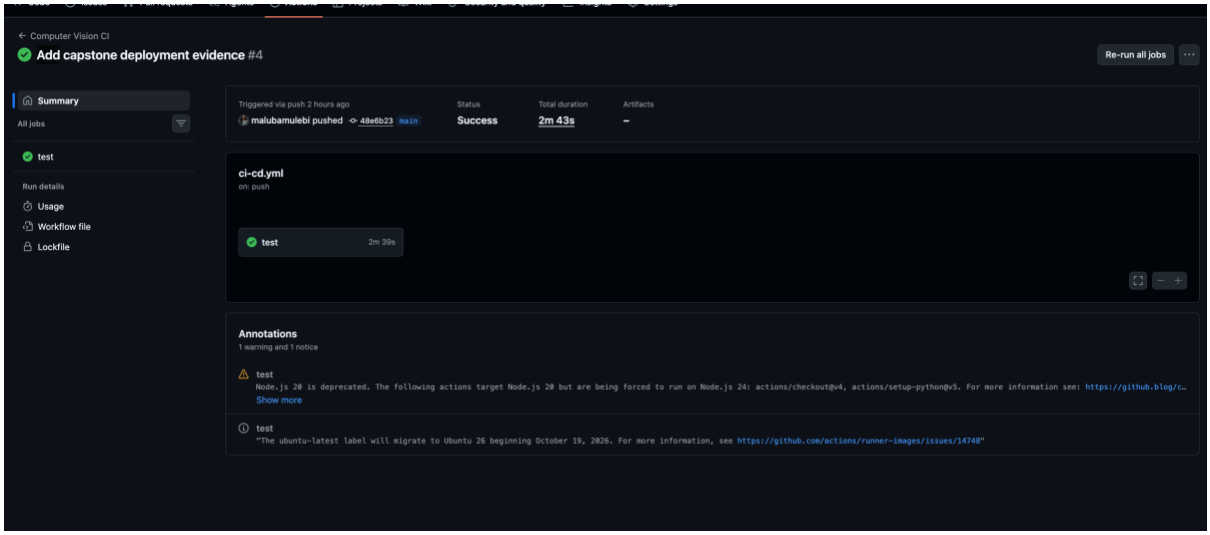


Figure 3: Successful GitHub Actions continuous integration workflow.

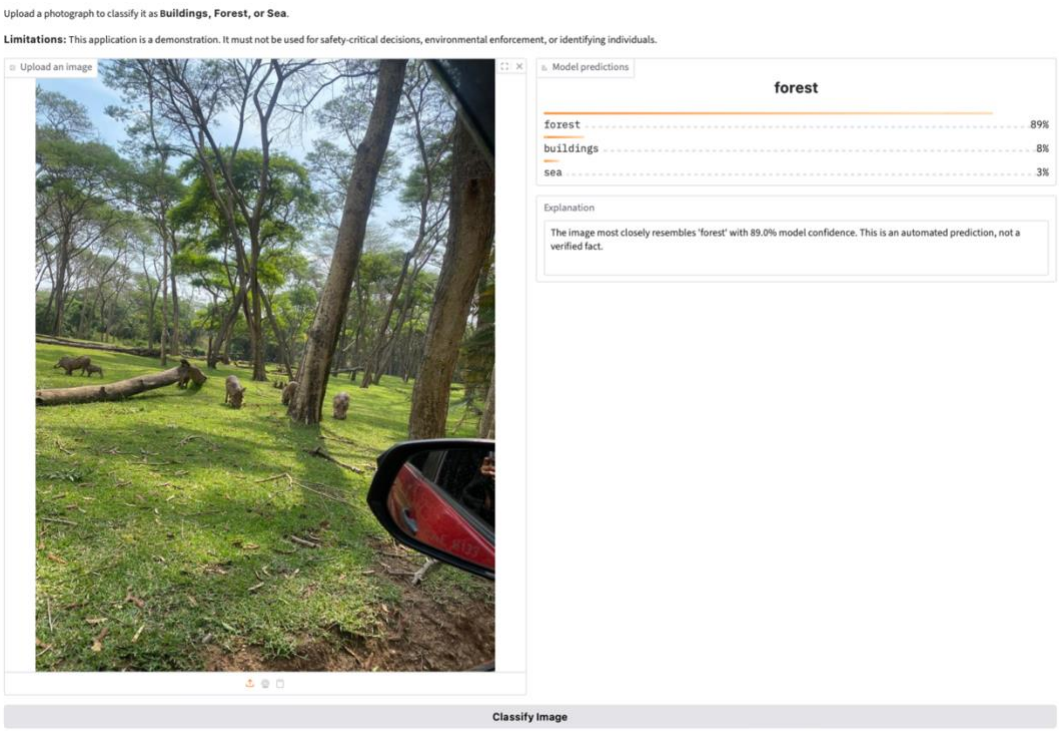


Figure 4: Gradio prototype showing forest classification with 89% confidence.

5. Ethics, Limitations and Impact Statement

Potential Application in Zambia

The classifier could support preliminary organisation of landscape photographs for environmental monitoring, educational research and image documentation in Zambia. However, it is not suitable for making independent environmental or regulatory decisions.

Potential Harm and Misuse

Incorrect classifications could lead to misleading records or inappropriate decisions. For example, classifying a forest image as a building scene could cause relevant vegetation information to be overlooked. False negatives may be particularly concerning in conservation applications because important environmental evidence could be missed.

Dataset Provenance and Consent

The system uses images from the Intel Image Classification dataset, reused from Practical 4. The dataset's original licence and usage permissions should be verified before commercial or operational deployment. The application does not require users to provide identifiable personal information, although uploaded photographs may unintentionally contain people or sensitive locations. Appropriate consent and privacy safeguards would therefore be necessary.

Bias and Generalisation

The model was trained using three image categories and may not represent Zambia's diverse environments adequately. Local landscapes, seasonal changes, lighting conditions and image

quality may influence prediction reliability. The reported 100% test accuracy does not establish equivalent performance on new Zambian photographs.

Mitigation and Responsible Deployment

The application displays confidence scores and an uncertainty warning for low-confidence predictions. Additional safeguards should include local dataset evaluation, human review of important decisions, privacy protection, regular performance monitoring and clear communication of system limitations.

Before operational deployment, the model would require validation using representative Zambian images, documented data permissions and an assessment of potential social and environmental consequences.

6. Conclusion

This capstone demonstrated the development, testing and deployment of a computer vision application using a fine-tuned MobileNetV3-Small model.

The classifier achieved 100% accuracy on the 135-image test set, while all seven automated tests passed. GitHub Actions successfully performed continuous integration, and Streamlit Community Cloud provided a publicly accessible application with automatic updates from GitHub.

Deployment challenges included Hugging Face hosting restrictions and Python dependency incompatibilities. These were addressed by selecting an alternative hosting platform and updating package versions.

Although the system performed well on the evaluated dataset, further testing with independent Zambian landscape images is necessary before considering real-world applications.

7. Individual Contribution and AI Assistance

This project was completed individually by Maluba Mulebi. The implementation reused the MobileNetV3 model developed during Practical 4.

AI assistance was used for code development, debugging, test preparation and documentation. The final application, evaluation results and automated tests were executed and verified by the student.

References

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